

TEMA

Topical Event Macroeconomic Assessment

An NLP-Based Framework for Mapping News Shocks
to India's Sectoral Outlook

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Problem Statement

Key Challenges in India's Macro - Economic & Trade Intelligence

Global Shocks + Information Overload

Global events and macroeconomic changes rapidly impact India's currency, inflation, and markets. At the same time, analysts are overwhelmed by huge volumes of repetitive news, making it difficult to find what actually matters.

Unstructured & Redundant News

News arrives in unstructured form with lots of noise and duplication. There is no proper categorization or clear mapping to Indian economic indicators, so extracting useful insights becomes slow and inefficient.

Solution : Tema Frame Work

Transforming unstructured news into structured macroeconomic intelligence

TEMA is a smart system that automatically collects news from 50+ News every day. It uses multiple layers of AI to filter out noise, understand what the news actually means, and show how it can impact the Indian economy — all in near real-time.

- **Multi-Layer Filtering** : Uses 3-step AI checking (keywords + LLM + meaning) to keep only relevant news with high accuracy.
- **Rich Taxonomy** : Organizes news into 6 main categories and 30+ sub-types directly linked to economic indicators.
- **Impact Intelligence** : Clearly shows whether the news is positive, negative, or neutral, which sector it affects, and key indicators to watch.
- **Production Ready** : Built with reliable technology (Django + local AI) and includes automatic error handling and logging.

Event Record

Trade Policy| Bangladesh Garment Export Tariff Cut

Date: 14 Apr 2025

Source: Reuters

Headline

Bangladesh cuts garment export tariff by 8% - effective June 2025

Classification

Event Class

Trade Policy

Sub-Type

Bilateral tariff change

Origin

Bangladesh

Direction

Negative (on India)

Sector & Impact

Sector

Textiles & Garments

Channel

Trade -> Supply Chain → Price

Trade Share

~4% of competing export

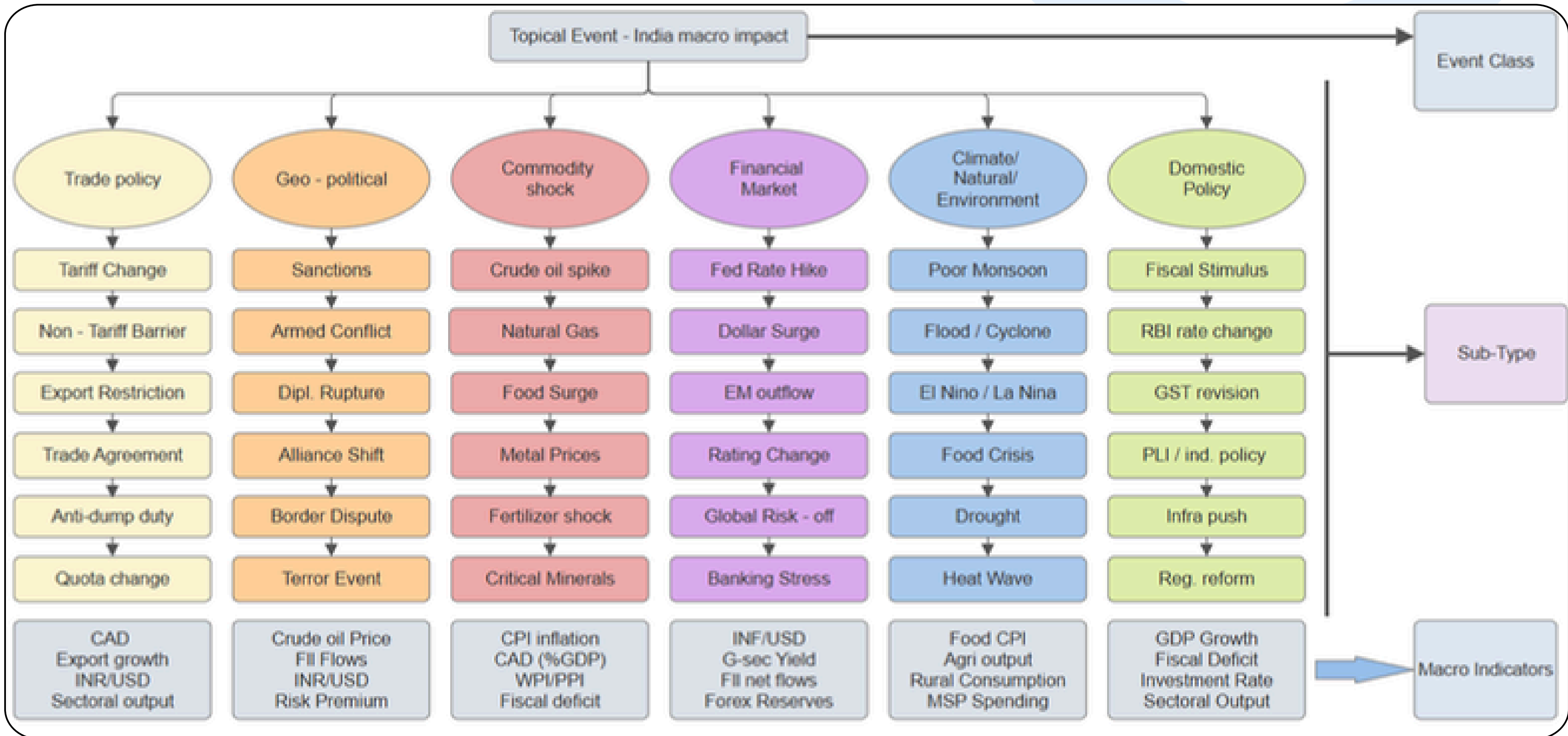
Key Indicator

Export growth - CAD - Output

Context & Implication

Bangladesh is India’s key competitor in global apparel markets. A lower export tariff raises their price competitiveness vs india in EU and US markets.

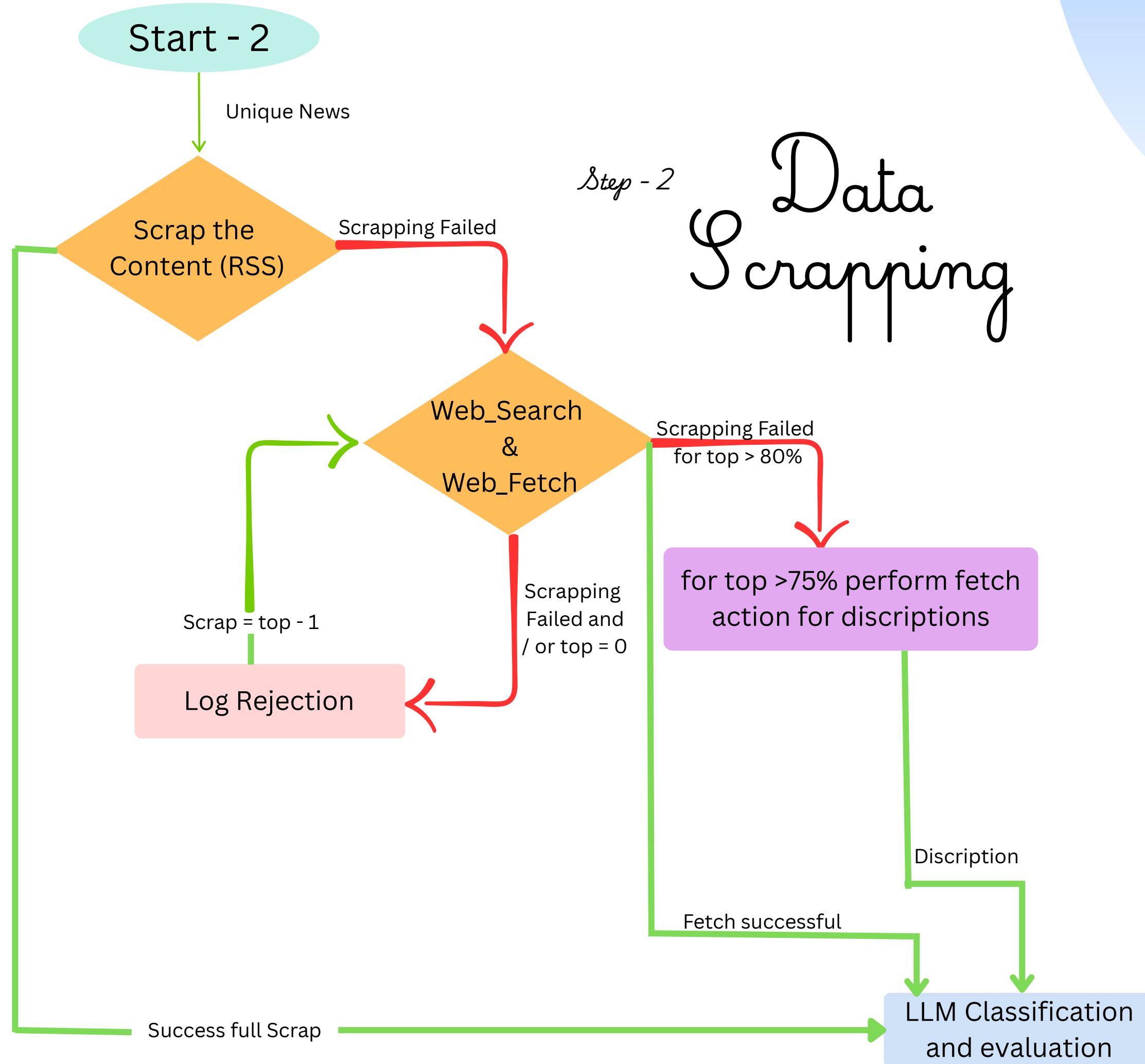
Topical Events





Data Cleaning Steps

1. Raw Feeds Fetching - Collect incoming news feeds from various RSS sources as the starting point the pipeline.
2. Keywords Pre-Filter - Feeds that do not match relevant keywords are immediately rejected and logged.
3. LLM Relevance Check - Use a LLM to evaluate contextual relevance. Irrelevant items are rejected and logged.
4. Semantic Duplication Check - Performs advanced semantic analysis to identify and remove duplicate news, ensuring uniqueness.
5. Logging & Rejection Management - All rejected feeds are properly logged for monitoring, auditing and future improvement of filters.

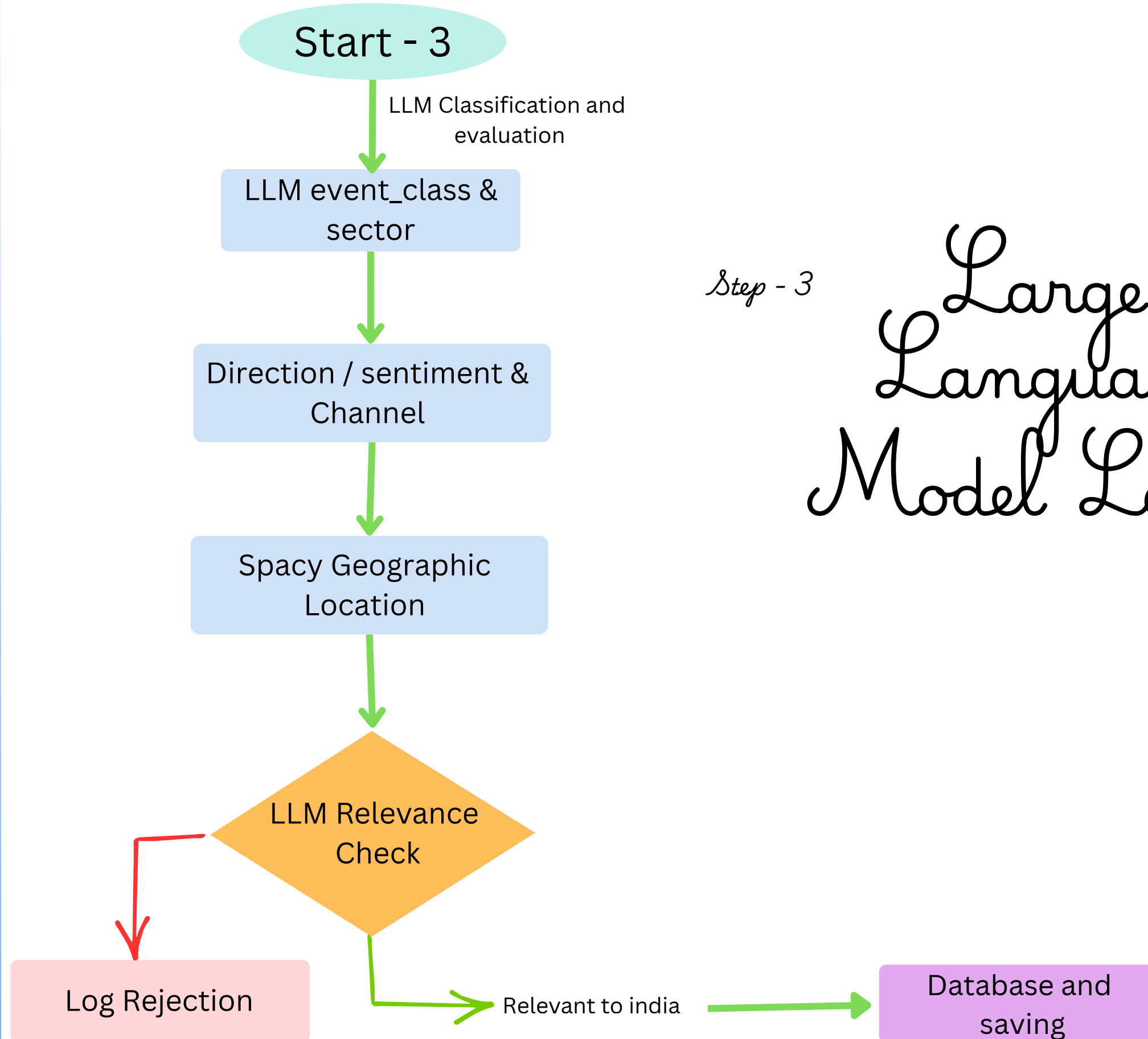


Data Scrapping Steps

1. Start → Unique News
2. Primary Scraping: RSS Content Scraping
3. Success → LLM Classification & Evaluation
4. Fail → Web Search + Web Fetch (Fallback)
5. Conditional Logic: If scraping fails for >80% items → Fetch descriptions for top 75% items
6. Failure Path: Log Rejection (for monitoring & retry)
7. Final Output: All successful content → LLM Classification and Evaluation.

Key Highlights:

- Multi-layer fallback for high reliability
- Smart conditional description fetching on high failure rate
- Clear rejection logging for debugging



Step - 3

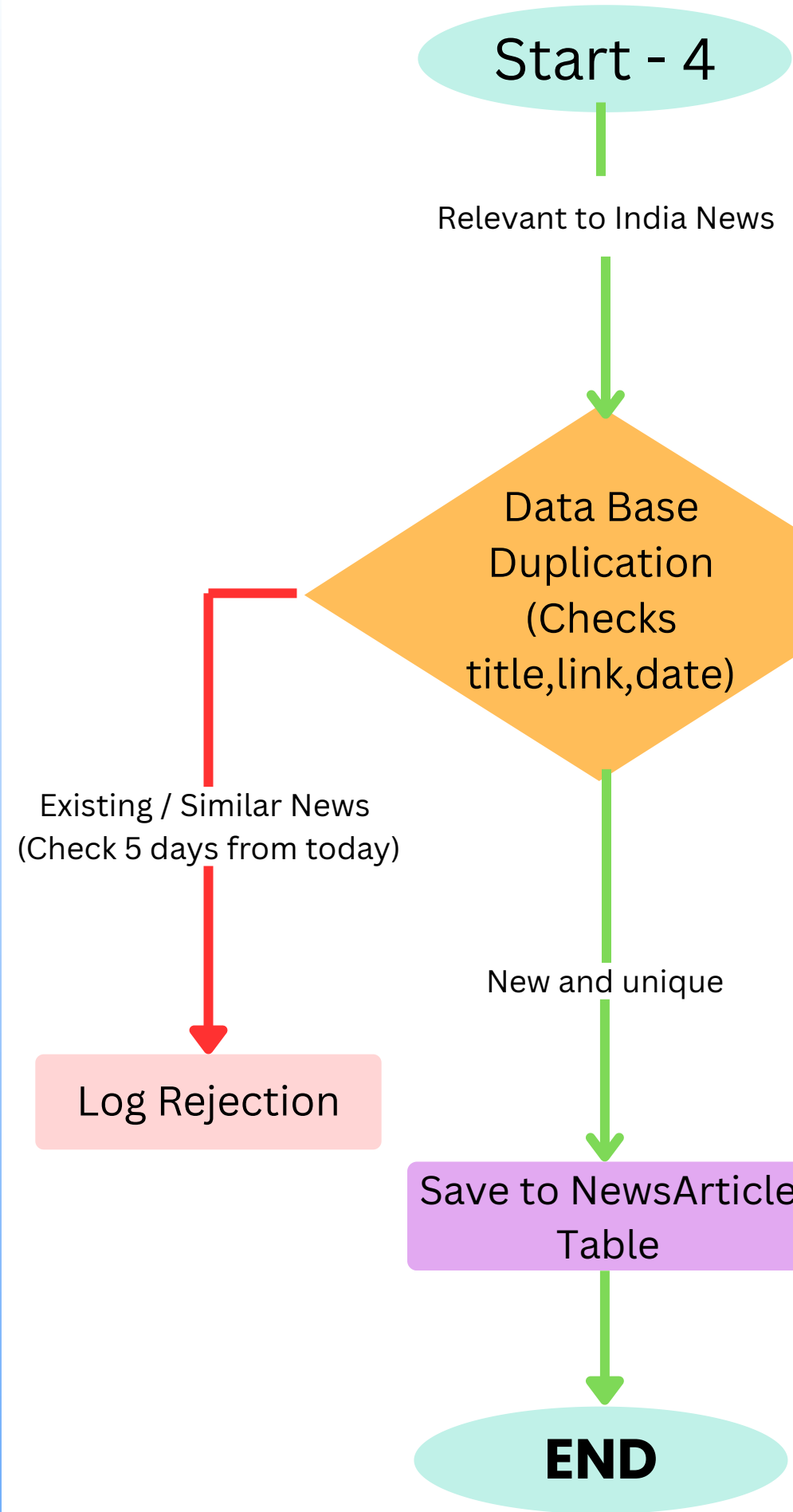
Large Language Model Layer

Large Language Model Layer

1. Start → LLM Classification & Evaluation
2. Stage 1: LLM → Event Classification + Sector Tagging
3. Stage 2: LLM → Direction / Sentiment + Channel Mapping
4. Stage 3: spaCy → Geographic Location Extraction
5. Stage 4: LLM Relevance Check
6. Not Relevant → Log Rejection
7. Relevant to India → Save to Database

Key Highlights:

- Multi-stage LLM pipeline for structured news intelligence
- Combines LLM reasoning + spaCy for location
- Final relevance filter ensures only India-focused news is stored



Step - 4

Data Base And its Management

Data Base And its Management

1. Start → Relevant to India News (from Step 3)
2. Database Duplication Check (title + link + date)
3. Existing / Similar News (within last 5 days) → Log Rejection
4. New & Unique → Save to NewsArticle Table
5. End

Key Highlights:

- Prevents duplicate news entries in the database
- Deduplication window: Last 5 days
- Only unique, India-relevant news gets stored

Start

Step 1 - Data Cheaning

- Data Cleaning is important to reduce Noise and Redundancy in the Data.
- This layer ensures that our pipeline is optimised and efficient by reducing (space and time complexity)

Step 2 - Data Scrapping

- Data Scrapping reduce chances of model hallucination, while in 3rd step i.e. LLM classification evaluation.
- Since we can get block there is a loop and alternate method to avoid hallucination.

Step 3 - Large Language Model Layer

- In this step we input clean data and from its title, discription, scrapped contents and publication date we try to generate insight direction and effect/impact of that perticulr information on indian economy.
- 2nd relevance test to avoid irrelevant content in DB.

Step 4 - Data Saving in the DB

- We check if the data is already present, if it is we reject / remove the data.
- If data is unique we save it to DB.
- To improve efficiency instead of searching whole DB for deduplication we Refer past 5 days data in the DB before saving it

END

Core Tech Stack

Architecture with privacy - first Artificial Intelligence

Backend & Data

- Django (Python)
- Django ORM + SQLite
- Json and CSV DB (train - test)

AI & NLP Layer

- Ollama + gemma3:4B (Local LLM)
- spacy NER for geolocation
- Relevance & Classification Prompts

Data Pipeline

- Feedparser (3 sources with 300+ Feeds)
- requests + beautifulsoup4 (bs4) for Scrapping
- Auto - retry + backoff on failure (scrapping loop).
- Semantic deduplications



All components designed for production reliability, data privacy (local LLM), and scalability.

Oil Jumps as Iran and Israel Exchange Fire With Truce at Risk



Date	07 Jun 2026	Source	Bloomberg Markets
Headline	Oil Jumps as Iran and Israel Exchange Fire With Truce at Risk		
Event class	Commodity Shock	Sub-type	Geopolitical Oil Supply Shock (Middle East Conflict, Oil Prices)
City/Country	Iran	Direction	Negative
Sector	Oil Gas and Refining	Channel	Geopolitical Oil Supply Shock & Commodity Prices & Imported Inflation & Current Account Deficit & INR Pressure
Insight	The exchange of fire between Iran and Israel, coupled with stalled ceasefire talks, has triggered a surge in oil prices. This disruption in the Middle East raises concerns about the stability of global energy supplies. The situation underscores the vulnerability of commodity markets to geopolitical instability.		

High Impact
Topical Events

India's economy is highly susceptible to global oil price shocks. Rising oil prices directly increase India's import costs, worsening its trade deficit and fueling inflation. This will force the RBI to likely raise interest rates to combat inflation, which will slow economic growth. The corporate sector, particularly energy-intensive industries, will face reduced profitability. The geopolitical uncertainty will also negatively impact investor sentiment and financial market stability, making this a decidedly negative development for India's economic outlook.

US notifies \$428 million sale of support services for Apache helicopters, M777A2 howitzers to India



Date	22 Jun 2026	Source	Hindustan Times
Headline	US notifies \$428 million sale of support services for Apache helicopters, M777A2 howitzers to India		
Event class	Trade Policy	Sub-type	Trade Defence (US-India Defence Sales, Defence Framework)
City/Country	India / US	Direction	Positive ⓘ
Sector	Defense and Aerospace	Channel	Trade competitiveness & Current account balance & INR exchange rate ⓘ
Insight	The United States has notified Congress of a \$428 million sale of support services to India, encompassing logistics, personnel training, and repairs for existing Apache helicopters and M777A2 howitzers. This transaction aims to sustain India's defense platforms and bolster its capabilities to address current and future threats, particularly within the Indo-Pacific and South Asia regions. Following the delivery of 28 Apache helicopters and 145 M777 Howitzers, contracted in 2015 and 2016 respectively, the sale underscores a strengthened U.S.-Indian strategic relationship and reflects ongoing efforts to deepen bilateral security ties, including high-level defense visits and a 10-year framework agreement. This sale is expected to proceed unless Congress objects, representing a significant investment in India's defense capabilities despite broader geopolitical tensions.		

Medium Impact Topical Events

The sale of support services will bolster India's defense capabilities, potentially reducing reliance on foreign maintenance and increasing domestic defense industry support, contributing to economic growth.

UK foreign secretary Yvette Cooper to meet Jaishankar to review ties



Low Impact Topical Events

Date	03 Jun 2026	Source	Hindustan Times
Headline	UK foreign secretary Yvette Cooper to meet Jaishankar to review ties		
Event class	Trade Policy	Sub-type	Bilateral Trade Relations (UK-India, Trade Agreements)
City/Country	India / China	Direction	Neutral
Sector	Foreign Trade and Exports	Channel	Trade competitiveness & INR exchange rate
Insight	UK Foreign Secretary Yvette Cooper's visit to India, following a similar trip by the UK's business and trade secretary, signals continued diplomatic engagement between the two nations. The meetings are focused on reviewing existing ties and potentially exploring new avenues for collaboration. This visit underscores the ongoing importance of bilateral relations between the UK and India.		

The news itself – a diplomatic review – doesn't inherently drive significant economic change. While it reinforces India's strategic position and could open doors for trade discussions, the content of those discussions remains unknown. Therefore, the impact on India's economy, financial markets, corporate sectors, and the RBI's policies is likely to be neutral. The RBI will continue its focus on inflation and growth, and the markets will react to broader global trends rather than this specific diplomatic event. Any positive or negative impact will depend entirely on the outcomes of the meetings and subsequent policy adjustments.

Case Study of Impact

Real - world news processed - High, Medium & Low Impact Case Study

High Impact

Event: Commodity Shock|
Sector: Oil & Gas
Direction : Negative
Insight: Rising oil prices increase India's import bill, worsen trade deficit, fuel inflation and force government to raise rates - slowing growth and pressuring INR

Medium Impact

Event: Trade Policy (Defense)|
Sector: Defense & Aerospace
Direction : Positive
Insight: \$428M Apache & M777 support services deal strengthens India - US strategic partnership and boosts domestic defense industry growth.

Low Impact

Event: Trade Policy (Bilateral)|
Direction : Neutral
Insight: Diplomatic Engagement is positive for long - term relations but does not immediately move markets or macro indicators. System correctly classified as low immediate economic impact.

Pipeline Performance Analysis

Overall Score: 7.2 / 10

The automated pipeline is functioning reasonably well. Some areas need improvement for higher reliability.

Performance Breakdown

Component	Score	Current Status
Summary Quality	8.5/10	Summaries are mostly accurate and well condensed.
India Impact Reasoning	6.8/10	Sometimes connects news to RBI/CAD/INR incorrectly.
Impact Score Accuracy	7.0/10	Direction is usually correct, but intensity can be off.
Classification Quality	6.5/10	Event class is okay, but sub-type and channel often mismatch.
Factual Grounding	7.5/10	Mostly stays close to source content.

Key Improvement Areas

1. Improve economic logic when linking news to RBI, inflation, CAD or INR.
2. Refine classification rules for sub-type and channel to reduce mismatches.
3. Add validation step to check if generated insight matches actual article content.

Key Challenges & Learnings

1. **Keywords Alone Are Not Enough:** Good for initial filtering, but full article content is mandatory for accurate insight generation.
2. **LLM Hallucinations on Channels:** Early versions produced inconsistent 'Channel' values. RAG + historical examples significantly improve consistency.
3. **Scraping is Fragile:** News sites frequently block or change structure. Built robust retry logic + fallback to web_search.
4. **Impact Scoring Needs Context:** Same event can have different impact depending on timing and India's current macro position (e.g., high vs low oil inventory).

Future Roadmap & Accuracy Improvements

Better News Data Sources

Move from free RSS to professional feeds (Bloomberg, Reuters Terminal, Event Registry) for lower latency and richer metadata.

Larger & Smarter Models

Upgrade from gemma3:4b to Llama 70B / Claude 3.5 / DeepSeek-R1. Use small fast model first, then route important news to big model.

Chain-of-Thought + Structured Output

Force LLM to output reasoning first, then structured JSON. Use Pydantic + few-shot examples for consistent fields (impact score -5 to +5, policy implication, affected indicators).

RAG + Historical Memory

Embed past news + correct Channels/Insights in vector DB. Retrieve similar historical cases before generating new insight → dramatically reduces hallucinations and improves Channel accuracy.



Thank You

Questions & Discussion

TEMA — Turning News into Macroeconomic Intelligence

Presented by Surabhi Yadav

Conclusion & Value Delivered

- TEMA successfully transforms unstructured global news into structured, India-specific macroeconomic intelligence.
- Multi-layer pipeline (Keyword → LLM → Semantic) achieves >85% relevance accuracy while maintaining low cost and full data privacy via local LLM.
- Real-world examples demonstrate the system's ability to correctly classify High, Medium, and Low impact events with clear sectoral channels.
- The framework is production-ready and scalable — with a clear roadmap to RAG-powered, context-aware decision intelligence.
- This project showcases end-to-end capability in building AI systems that solve real economic information problems.